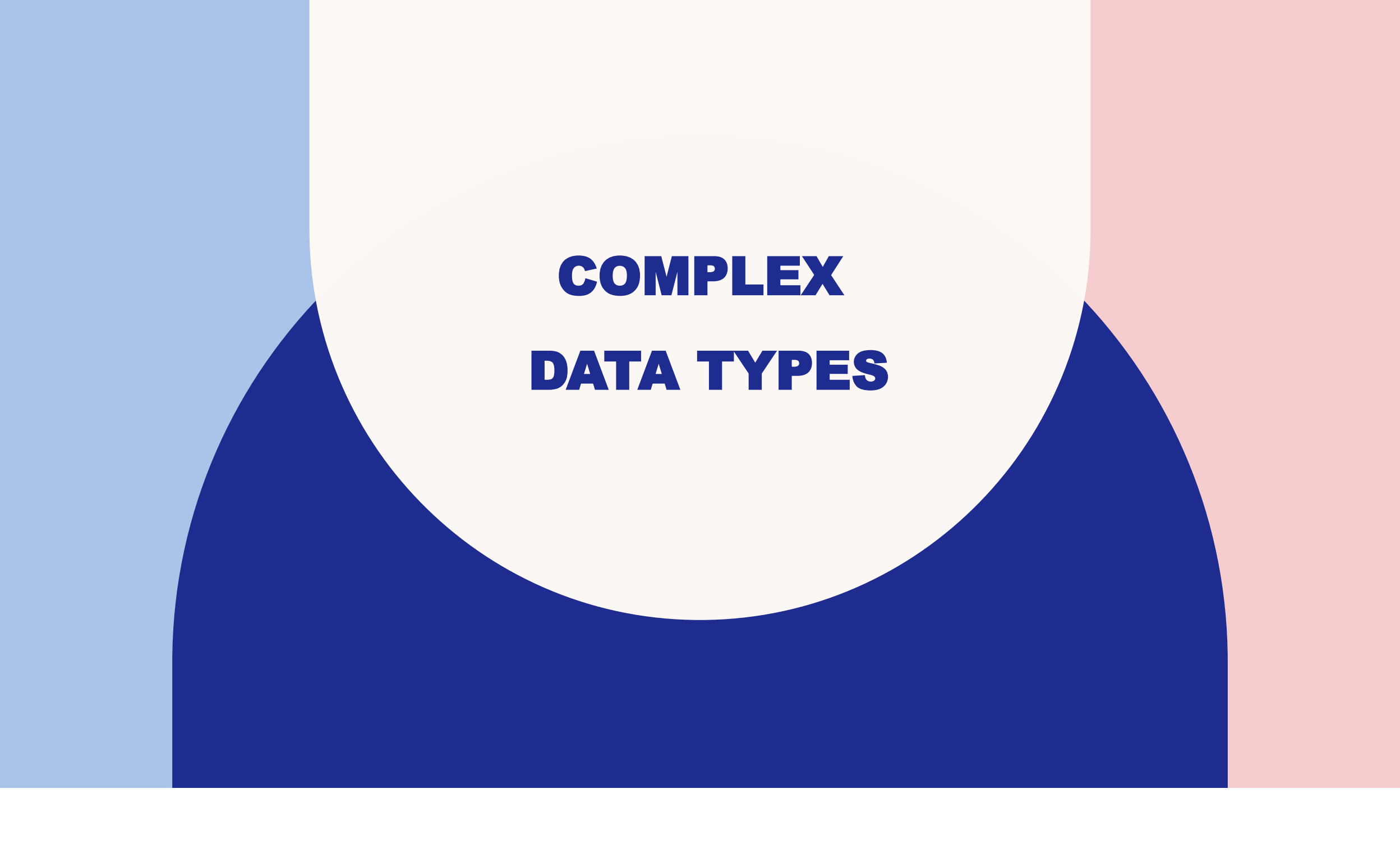


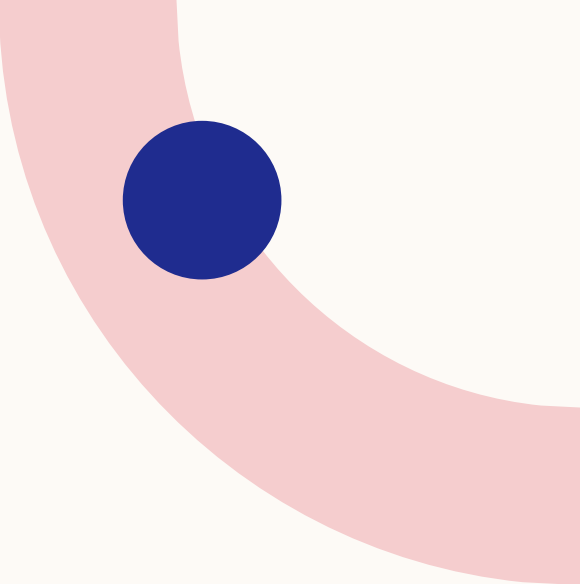
The background features a large white circle in the center, partially overlapping a light blue rectangle on the left and a light pink rectangle on the right. A dark blue shape, resembling a wide, shallow bowl or a large arc, sits at the bottom, partially covering the white circle and the other background elements.

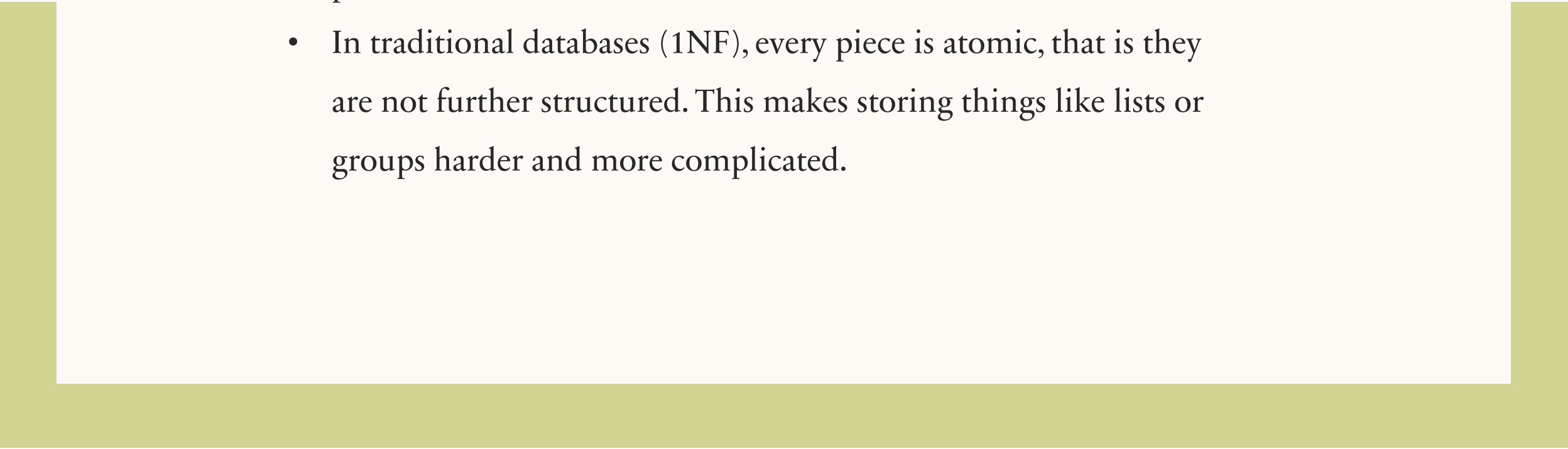
COMPLEX DATA TYPES

COMPLEX DATA TYPES

- INTRODUCTION
- WHY COMPLEX DATA TYPES
- WHAT ARE COMPLEX DATA TYPES
- EXAMPLE
- CONCLUSION

INTRODUCTION

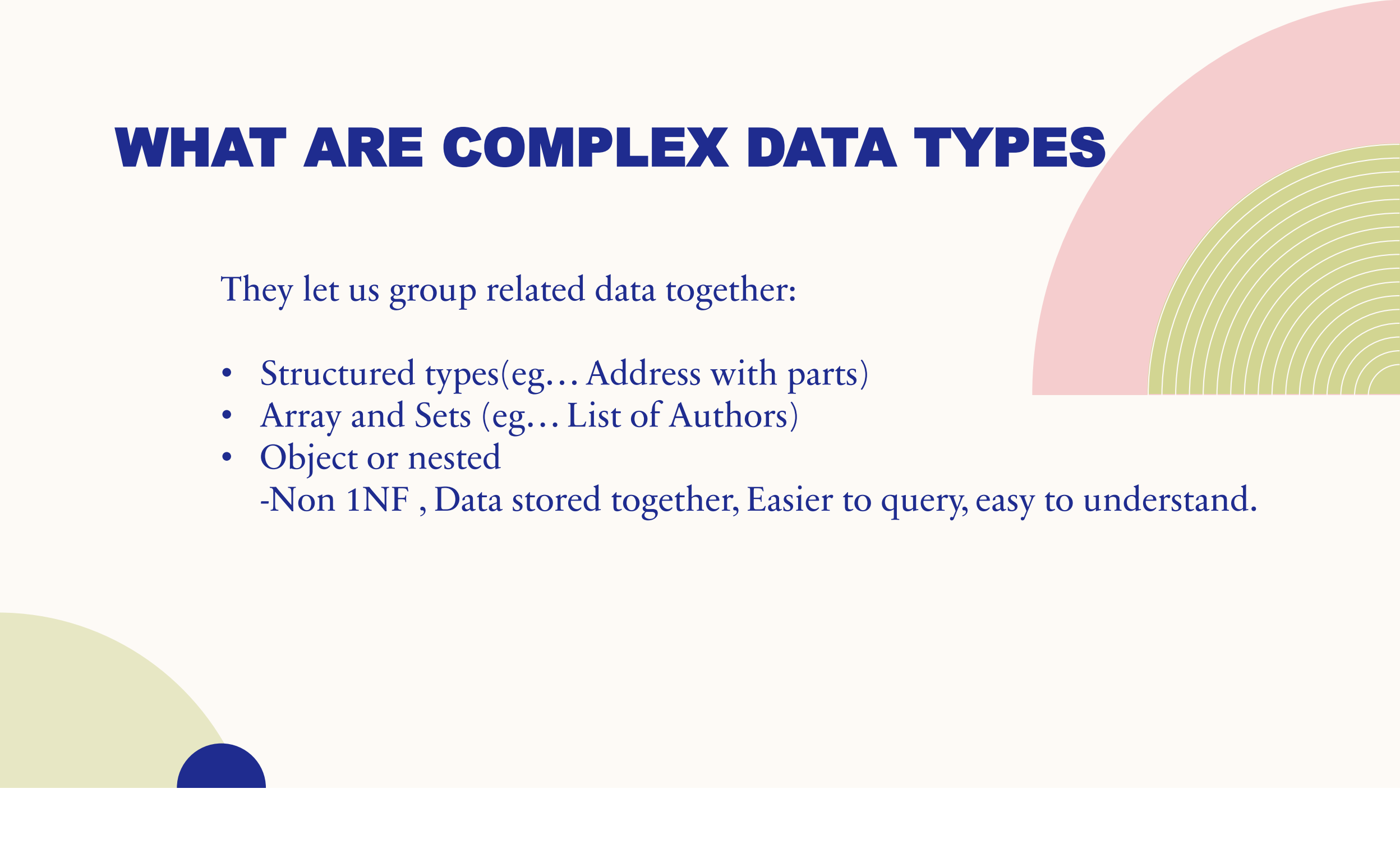


- Traditional databases use simple data types (like numbers and text). But real-world data can be more complex, like multiple phone numbers or detailed addresses.
 - In traditional databases (1NF), every piece is atomic, that is they are not further structured. This makes storing things like lists or groups harder and more complicated.
- 

WHY COMPLEX DATA TYPES

- To make it easier to represent real-life data in databases, like:
 - Addresses with street, city, and pin code.
 - A book with multiple authors.
 - A student with many phone numbers.
- Complex Type Systems can represent E-R model concepts, such as composite attributes, multivalued attributes, generalization, and specialization directly, without a complex translation to the relational model.

WHAT ARE COMPLEX DATA TYPES



They let us group related data together:

- Structured types(eg... Address with parts)
- Array and Sets (eg... List of Authors)
- Object or nested
 - Non 1NF , Data stored together, Easier to query, easy to understand.

EXAMPLE

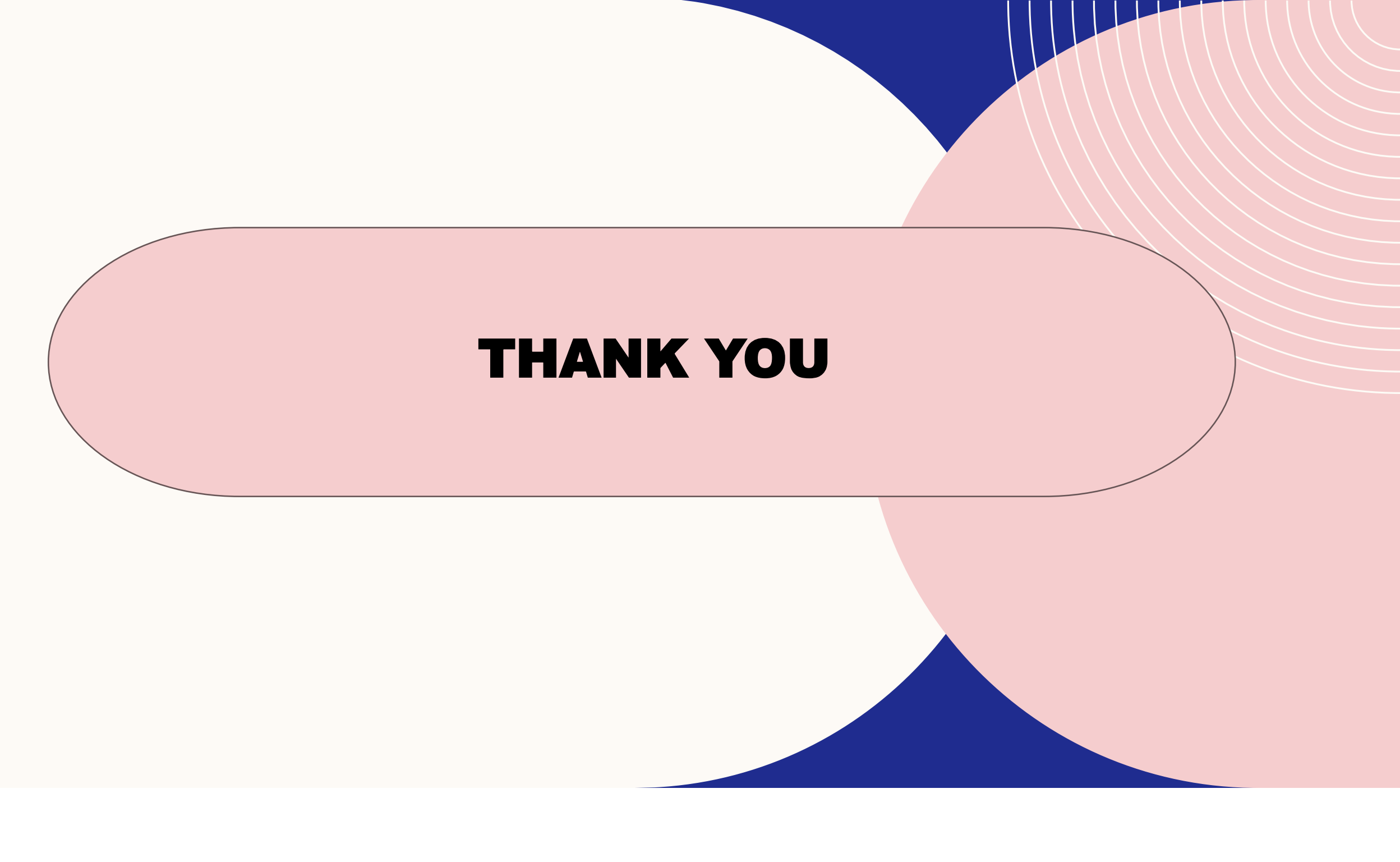
A book may have:

- - Title
- - List of authors(Array)
- - Publisher
- - Keywords

These can be stored easily using complex types.

CONCLUSION

- Complex types are powerful, but can cause redundancy if not used carefully (e.g., storing the same relation in two places)
- Complex data types make it easier to represent real-world information. They are great for object-oriented design, APIs, and modern applications.



THANK YOU